

UCO Collection Business

Mumbai scale decision brief | 5 October 2026 | Decision focus: prove collection economics before investing in processing

Recommendation

Start with collection and aggregation; defer manufacturing. Validate a dense Mumbai cluster, secure a written buyer formula, and demonstrate repeat paid deliveries. At the PDF plan’s ₹50/L buy and ₹62/L sell assumptions, a 1 lakh L/month hub can show about ₹4.14 lakh monthly operating profit before financing, depreciation and tax. The result is highly price-sensitive and needs about ₹40 lakh of cash-cycle funding.

Price evidence: do not mix unlike prices

Figure	What was published	Use in decision
₹75/L UCO purchase	Reported Geekay Bio Products transaction, Coimbatore, Sep 2025. Source report	Historical transaction reference only; not a current Mumbai quote.
₹80/L finished biodiesel	OMC procurement benchmark for Apr-Jul 2025, excluding GST and transport. Source report	Finished fuel benchmark, not collected UCO; tender terms and period matter.
14.1% EBITDA; 8.6% net	Rajputana Biodiesel FY25: income ₹68.6 crore, EBITDA ₹9.64 crore, profit ₹5.92 crore. Annual report	Manufacturing context only; business includes several feedstocks/products.
₹50/L buy; ₹62/L sell	Assumptions in the supplied plan.	Model inputs only until confirmed by written, current buyer offers.

Decision gates

Proceed with an 8-week collection pilot only after two indicative buyer offers, required current enrolment/authorisation evidence, measured supply density and a feasible site/permissions check. Scale only after realised contribution is at least ₹6/L, losses are at most 3%, receivables are at most 14 days and no single buyer controls the route.

The ₹80/L finished biodiesel benchmark and ₹75/L UCO report are historical, geographically distinct public figures. They do not establish today’s spread. All prices need fresh counterpart confirmation.

Collection economics at scale

Illustrative monthly model using the plan's ₹50/L purchase and ₹62/L sale inputs. Assumes 3% rejected/lost volume; hired transport; prices exclude GST. Figures are estimates, not quotes.

Monthly litres	30,000 pilot	1 lakh hub	3 lakh network
Revenue after 3% loss	₹18.04L	₹60.14L	₹180.42L
Oil purchases	₹15.00L	₹50.00L	₹150.00L
Collection, transport, handling	₹1.80L	₹4.00L	₹10.50L
Fixed overhead	₹1.00L	₹2.00L	₹5.00L
Operating profit before finance, depreciation, tax	₹0.24L	₹4.14L	₹14.92L
Operating margin	1.3%	6.9%	8.3%

Sensitivity at 1 lakh L/month

Change from base case	Approx. monthly operating profit	What it means
Base: buy ₹50/L, sell ₹62/L	₹4.14L	Before interest, depreciation and tax.
Selling price falls ₹5/L	-₹0.71L	A ₹5 change erases the base profit.
Buying price rises ₹5/L	-₹0.86L	Do not lock supplier prices without matched offtake pricing.
Target ₹4/L operating profit	Max buy ≈₹50.14/L	At ₹62/L realised selling, with 3% loss and ₹6/L variable + fixed cost allocation.

Working capital and cash cycle

At 1 lakh L/month and the base assumptions: 7 days of inventory costs about ₹11.7L (₹50/L purchase cost); 14 days of receivables tie up about ₹28.1L (realised revenue after loss); suppliers are paid same day. Total cash tied up is about ₹39.7L, before operating reserve. Budget ₹40-45L for working capital plus ₹8-12L ramp-up reserve. Customer credit or slower collections increase this need directly.

Supply density check: at 250 L discarded/site/month, 1 lakh L needs about 400 sites; at 1,000 L/site/month, about 100 sites. The plan's 15-20 kitchens are suitable for a pilot, not this hub scale. Measure discarded oil actually available.

Collection hub: itemised capital

Design basis: 1 lakh L/month, about 4,000 L per working day, rented 1,500-2,500 sq ft industrial premises, hired transport and approximately 25,000 L segregated tank capacity.

Item	Scope / quantity	Capital (₹ lakh)	Evidence / status
Storage/settling tanks	5 × 5,000 L stainless steel	19.0	Published ₹3.8L/tank benchmark; suitability and current price to confirm.
Filter press	One, advertised ~5-7 t/day	2.5-3.5	Supplier advertised range. Filtration does not remove dissolved water/FFA.
Transfer pumps	2 × 2 HP pumps	1.5	Published ₹75,000/pump benchmark; hoses budgeted below.
Collection drums	600 × 50 L numbered HDPE	1.98	Published ₹330 each; refresh quote.
Scales, sampling, basic tests	Hub weighing and incoming QC	1-2	Planning estimate.
Premises deposit and fit-out	Rented industrial unit	3-5	Estimate; depends on location and lease.
Spill containment, fire, electrical, hoses	Site-specific installation	3-4	Estimate; final design/authority requirements vary.
Permissions, records, setup	Documentation and traceability	1-2	Estimate. Determine requirements for exact process/site.
Freight, installation, contingency	Allowance	4-5	Estimate.
Setup subtotal	Before working capital	38-44	Rounded from itemised range.
Working capital + reserve	7 inventory / 14 AR days; reserve	48-57	₹40-45L cash cycle + ₹8-12L reserve.
Total funding envelope	Hired transport; before tax/ITC timing	86-101	Tax basis of supplier prices unconfirmed; add applicable tax and timing buffer.

Published anchors

[Sundrex prospectus](#) disclosed ₹3.8L/5,000 L SS tank and ₹75,000/2 HP pump supplier pricing through July 2025. [Sri Samy Engineering](#) advertised a ₹2.5-3.5L filter press. [Om Containers](#) advertised ₹330/50 L drum. Prices, taxes, freight and suitability require written current quotes.

A lower cost configuration may be possible with buyer-approved tanks and direct dispatch. For comparison, Tata's Sep 2025 announced Ace Gold+ price was ₹5.52L ex-showroom; a road-ready owned vehicle costs more. Hired transport is assumed here. [Tata launch release](#)

Own biodiesel plant: conditional case

A plant is a separate investment. At 20,000 L/day and 26 operating days/month, 95% volumetric yield implies feedstock demand of about 5.5 lakh L/month. A 1 lakh L collection hub covers roughly 18% of that requirement.

Indicative fixed investment at 20,000 L/day

Package	Estimated capital (₹ crore)	Scope to quote
Feedstock/fuel/chemical tanks; unloading	0.50-0.80	Segregated storage, unloading, bunding.
Filtration, dehydration, high-FFA pretreatment	0.60-1.00	Feedstock variability drives pretreatment scope.
Reaction, separation, methanol recovery	1.30-2.00	Core process package and controls.
Fuel purification/drying; glycerol handling	0.50-0.80	Product spec and by-product route.
Boiler/heater, cooling, power/utilities	0.50-0.80	Utility loads and redundancy.
Effluent treatment, fire, chemical safety	0.30-0.50	Site/process specific design.
Quality lab and instruments	0.15-0.30	Fuel quality release testing.
Civil works, piping, installation	0.65-0.80	Excludes land.
Engineering, commissioning, contingency	0.50-1.00	Performance testing and contingency.
Fixed investment	5.00-8.00	Excludes land and applicable taxes.
Working capital + operating reserve	3.00-4.00	Feedstock, chemicals, receivables, ramp-up.
Total funding envelope	8.00-12.00	Tax/ITC timing may raise peak funding.

Cost source and economics

Magtech published ₹5 crore for a 20,000 L/day plant in Apr 2024, excluding land. That is a supplier estimate, not a turnkey quote. The item allocations above are planning estimates that sum to ₹5-8 crore fixed investment. [Published supplier estimate](#)

At historical ₹80/L finished fuel, ₹62/L feedstock, assumed ₹12/L conversion cost and ₹1/L by-product credit, indicative contribution is only ₹3.74/L before fixed overhead, finance and depreciation. At ₹75/L feedstock it is negative. Get a toll-processing trial and buyer specification first; confirm yield, quality deductions and by-product realisation.

Plant prerequisites before ordering

Obtain feedstock tests across seasons; process design and vendor performance guarantees; site-specific MPCB, fire, chemical storage and effluent approvals; fuel buyer qualification and written specification; methanol and glycerol handling plans; confirmed utilities, wastewater route, commissioning acceptance test, spare parts and operator training. No plant order should precede supply and offtake proof.

Evidence limits

The capex split is an estimated budgeting framework, not a vendor bill of materials. The ₹80/L OMC figure is a historical procurement benchmark and may not represent a current tender, delivery basis or eligible supplier price.

People and counterparties to approach

Use these public channels to identify the procurement, operations or engineering decision-maker. These are prospects and contact routes, not verified offers or confirmed willingness to transact. No one has been contacted.

Need	Organisation / person to request	Public contact
UCO buyer / aggregator	Muenzer Bharat, Navi Mumbai: UCO Procurement & Business Development	+91 93724 29729 office@muenzer.in Contact page
UCO buyer / collection partner	ARIS BioEnergy: procurement or aggregator partnerships	+91 73047 45554 frontoffice@arisbioenergy.com Contact page
Kitchen supply source	Prashant Corner, Thane: central production/procurement head	+91 70459 30190 customer.care@prashantcorner.com Contact page
Kitchen supply source	Jhama Sweets, Mumbai: operations/factory manager	+91 82602 24466 customercare@jhamasweets.com Contact page
Multi-outlet collection	Jumboking: supply-chain/operations head	022-42665014 customersupport@jumboking.co.in Contact page
Institutional kitchens	Sodexo: MasterKitchen / corporate responsibility	gagan.vaibhav@sodexo.com luna.mishra@sodexo.com Contact page
Plant quotation	Magtech: sales/engineering	+91 6289715123 sales@magellanium.com Published estimate
Future fuel sales	IOCL, BPCL, HPCL: procurement/tender teams	Institutional route for finished biodiesel; qualify against current tender and fuel specification before investing.

What to ask each prospective buyer

Request current ₹/kg offer and validity; accepted moisture, insolubles, FFA and other limits; sampling method and rejection/deduction schedule; delivery point and minimum load; freight responsibility; payment days; required enrolment/authorisation; weighing basis and density conversion. Ask whether price floats with a published index or tender. Do not use an indicative phone price as the model's realised price.

Sources and regulatory checks

FSSAI's 2022 order addresses RUCO enrolment/authorisation requirements: [FSSAI order](#). Its guidance explains used cooking oil controls; the 50 L/day fresh-oil threshold concerns recordkeeping and does not exempt smaller kitchens from safe-oil requirements: [FSSAI guidance note](#). Confirm applicability and current status directly before launch. Check site/process requirements with [MPCB consent guidance](#).

All figures in this brief are INR, with lakh/crore notation. Public prices are dated historical observations or advertised prices; estimates are labelled. Obtain current written quotes and professional site-specific compliance advice.

Step-by-step launch guide

Follow the gates in order. Indicative weeks are planning windows, not promised authority timelines. Do not commit to manufacturing capex until the collection business and fuel route are proven.

1. Choose one dense collection cluster and set the budget. Pick an area with short routes and a workable rented hub. Set a pilot cash cap, hiring plan and decision owner. Output: one-page cluster map and pilot budget.
2. Get two buyer indications and confirm current NFP/RUCO evidence. Approach at least two enrolled non-food production units or their authorised commercial teams for written indicative terms in ₹/kg, quality limits, deductions, minimum delivery and payment days. Verify current enrolment/authorisation evidence before buying equipment. Output: comparable buyer term sheet and compliance checklist.
3. Survey 30-50 kitchens and weigh actual discarded UCO. Record site, weekly litres, pickup frequency, storage, contamination and decision-maker. Observe and weigh real discarded oil; do not infer UCO from fresh-oil use. Output: route-level litres/week and supplier pipeline.
4. Test samples and agree the pricing basis. Send representative samples to buyer; record acceptance, moisture/impurity/FFA feedback and deductions. Agree price per kg and weigh method; if modelling litres, confirm density and conversion. Output: accepted sample and net realised-price formula.
5. Establish the business and clear site/process requirements. Set up entity, current account and accountant review of GST applicability. Confirm buyer authorisation; obtain site-specific MPCB, fire, waste and storage determinations before signing lease or processing. Output: written requirement list and feasible site.
6. Sign supplier and offtake terms. Match buying price reviews to buyer pricing, spell out quality, weighing, rejected loads, pickup and payment dates. Avoid fixed supply prices against floating sales. Output: signed terms with workable spread and short cash cycle.
7. Run the pilot with hired transport and traceable containers. Use numbered, sealed containers; weigh and sample each batch; record source, date, gross/tare/net weight, quality, pickup and buyer receipt. Keep inventory segregated. Output: clean batch history and weekly route cost.
8. Complete an 8-week paid-delivery pilot and apply the gate. Track litres, rejection/loss, contribution after variable costs, route density, payment time and buyer concentration. Scale only if loss $\leq 3\%$, contribution $\geq ₹6/L$, receivables ≤ 14 days and no single buyer dependence. Otherwise repair route, terms or price and repeat. Output: signed-off unit economics.
9. Fund and expand collection routes. Reserve at least the 7-day inventory + 14-day receivable cash need, plus operating reserve. Add routes in dense increments and monitor same metrics monthly. Output: reliable 1 lakh L/month supply plan with cash runway.
10. Trial toll conversion only if a finished-fuel route is credible. Send representative UCO to a toll processor. Confirm yield, quality, conversion cost, by-product credit, acceptance and buyer payment terms. Output: verified per-litre conversion economics and qualified buyer path.
11. Consider an owned plant only after proof. Commission detailed engineering, fixed-price itemised bids and performance guarantees; secure feedstock supply, site permits, utilities, effluent/fire/chemical controls and fuel-buyer qualification. Validate working capital and downside case. Output: investment case with binding supply/offtake, approved site and commissioning acceptance criteria.

Scale decision: collection first; plant only when both feedstock and offtake are contractable and downside economics still work.